

CA – 01

Examination

INT - 330

VPC CREATION AND LANUCHING THE LINUX SERVER

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Roll No. - 25

PROBLEM STATEMENT

A Startup wants to make its company website accessible to users across the internet. The website will run on a Linux Server and must allow HTTP access from any location. The Infrastructure should be deployed inside a custom network environment. What AWS Resources and configuration are required to achieve this? Perform implementation.

Step 01- Created the VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/4) Info

Find VPCs by attribute or tag

Last updated 1 minute ago Actions Create VPC

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
my-vpc-01	vpc-0469678595a1c1cbe	Available	-	-	Off	10.0.0.0/16
StartUp-VPC	vpc-0cf6c70159c65ea7f	Available	-	-	Off	10.0.0.0/16
-	vpc-0b53dba63d46e9fc4	Available	-	-	Off	172.16.0.0/16

vpc-0cf6c70159c65ea7f / StartUp-VPC

VPC

Your AWS virtual network

StartUp-VPC

Subnets (2)

Subnets within this VPC

us-east-1a

StartUp-Public-subnet

us-east-1b

StartUp-Private-Subnet

Route tables (4)

Route network traffic to resources

rtb-0c356dba953bce69

StartUp-Public-RouteTable

rtb-0ff830c13f9c73afe

StartUp-Private-RouteTable

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

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Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
my-vpc-01	vpc-0469678595a1c1cbe	Available	-	-	Off	10.0.0.0/16
StartUp-VPC	vpc-0cf6c70159c65ea7f	Available	-	-	Off	10.0.0.0/16
-	vpc-0b53dba63d46e9fc4	Available	-	-	Off	172.16.0.0/16

vpc-0cf6c70159c65ea7f / StartUp-VPC

Details Resource map CIDRs Flow logs Tags Integrations

Details

VPC ID vpc-0cf6c70159c65ea7f

State Available

DNS resolution Enabled

Main network ACL acl-0b0f3b20587acc6e3

IPv6 CIDR (Network border group) -

Tenancy default

Default VPC No

Network Address Usage metrics Disabled

Block Public Access Off

DHCP option set dopt-087912145d4aaefc7

IPv4 CIDR 10.0.0.0/16

Route 53 Resolver DNS Firewall rule groups

DNS hostnames Disabled

Main route table rtb-0ff830c13f9c73afe

IPv6 pool -

Owner ID 307619016552

CloudShell Feedback

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Step 2:- Create private and Public Subnets

Subnets (1/11) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	StartUp-Public-subnet	subnet-0edb5ba3f1876812d	Available	vpc-0cf6c70159c65ea7f Start...	Off	10.0.1.0/24
☐	-	subnet-0e42decc718bbd82f	Available	vpc-0b53dba63d46e9fc4	Off	172.31.32.0/2
☐	-	subnet-0fe39eb6b1c7b9bbf	Available	vpc-0b53dba63d46e9fc4	Off	172.31.16.0/2
☐	my-public-subnet-01	subnet-084a4d9cff312871c	Available	vpc-0469678595a1c1cbe my-...	Off	10.0.0.0/24
☐	my-private-subnet-01	subnet-0fb8a89f0d762fbb8	Available	vpc-0469678595a1c1cbe my-...	Off	10.0.2.0/24

subnet-002e3f6188c269f01 / StartUp-Private-Subnet

[Details](#) [Flow logs](#) [Route table](#) [Network ACL](#) [CIDR reservations](#) [Sharing](#) [Tags](#)

Details

Subnet ID
[subnet-002e3f6188c269f01](#)

IPv4 CIDR
[10.0.2.0/24](#)

Availability Zone
[use1-az1 \(us-east-1b\)](#)

Network ACL
[acl-0b0f3b20587acc6e3](#)

Subnet ARN
[arn:aws:ec2:us-east-1:30761901655:2:subnet/subnet-002e3f6188c269f01](#)

Available IPv4 addresses
[251](#)

Network border group
[us-east-1](#)

Default subnet

State
Available

IPv6 CIDR
-

VPC
[vpc-0cf6c70159c65ea7f](#) | StartUp-VPC

Auto-assign public IPv4 address
No

Block Public Access
Off

IPv6 CIDR association ID
-

Route table
[rtb-0d4f0bae70a1a7b57](#) | StartUp-Private-RouteTable

Auto-assign IPv6 address

Subnets (1/11) [Info](#)

Last updated 2 minutes ago [Actions](#) [Create subnet](#)

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐	-	subnet-0b3adac0e43cc7927	Available	vpc-0b53dba63d46e9fc4	Off	172.31.80.0/2
☒	StartUp-Public-subnet	subnet-0edb5ba3f1876812d	Available	vpc-0cf6c70159c65ea7f Start...	Off	10.0.1.0/24
☐	-	subnet-0e42decc718bbd82f	Available	vpc-0b53dba63d46e9fc4	Off	172.31.32.0/2
☐	-	subnet-0fe39eb6b1c7b9bbf	Available	vpc-0b53dba63d46e9fc4	Off	172.31.16.0/2
☐	my-public-subnet-01	subnet-084a4d9cff312871c	Available	vpc-0469678595a1c1cbe my-...	Off	10.0.0.0/24

subnet-0edb5ba3f1876812d / StartUp-Public-subnet

[Details](#) [Flow logs](#) [Route table](#) [Network ACL](#) [CIDR reservations](#) [Sharing](#) [Tags](#)

Details

Subnet ID
[subnet-0edb5ba3f1876812d](#)

IPv4 CIDR
[10.0.1.0/24](#)

Availability Zone
[use1-az6 \(us-east-1a\)](#)

Network ACL
[acl-0b0f3b20587acc6e3](#)

Subnet ARN
[arn:aws:ec2:us-east-1:30761901655:2:subnet/subnet-0edb5ba3f1876812d](#)

Available IPv4 addresses
[251](#)

Network border group
[us-east-1](#)

Default subnet

State
Available

IPv6 CIDR
-

VPC
[vpc-0cf6c70159c65ea7f](#) | StartUp-VPC

Auto-assign public IPv4 address
No

Block Public Access
Off

IPv6 CIDR association ID
-

Route table
[rtb-015c52ab93a1b8ad6](#) | StartUp-Public-RouteTable

Auto-assign IPv6 address

Step 3:- Created the Internet Gateway and NAT Gateway

The screenshot shows the AWS VPC console with the 'Internet gateways' page selected. The left sidebar contains the 'VPC dashboard' and a list of VPC resources including 'Virtual private cloud', 'Security', and 'PrivateLink and Lattice'. The main content area displays a table of Internet gateways. The 'StartUp-igw' gateway is selected, and its details are shown below the table.

Name	Internet gateway ID	State	VPC ID	Owner
Work IGW	igw-030ae43583753fd24	Attached	vpc-0245f8065100bccbc Work VPC	307619016552
-	igw-05a62871be273573e	Attached	vpc-0b53dba63d46e9fc4	307619016552
StartUp-igw	igw-09757064680bd0c44	Attached	vpc-0cf6c70159c65ea7f StartUp-VPC	307619016552
my-internet-gateway	igw-0be92084c67ac2bfa	Attached	vpc-0469678595a1c1cbe my-vpc-01	307619016552

igw-09757064680bd0c44 / StartUp-igw

Details

Internet gateway ID igw-09757064680bd0c44	State Attached	VPC ID vpc-0cf6c70159c65ea7f StartUp-VPC	Owner 307619016552
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The screenshot shows the AWS VPC console with the 'NAT gateways' page selected. The left sidebar contains the 'VPC dashboard' and a list of VPC resources including 'Virtual private cloud', 'Security', and 'PrivateLink and Lattice'. The main content area displays a table of NAT gateways. The 'StartUp-ngw' gateway is selected, and its details are shown below the table.

Name	NAT gateway ID	Connectivity...	State	State message	Availability ...	Route table ID	P
StartUp-ngw	nat-13980b9f202e1ac0a	Public	Available	-	Regional	rtb-0c356dbaa9...	1
my-nat-gateway--01	nat-1207c890805023957	Public	Available	-	Regional	rtb-0a20e0542...	4

nat-13980b9f202e1ac0a / StartUp-ngw

Details

NAT gateway ID nat-13980b9f202e1ac0a	Availability mode Regional	State Available	State message -
NAT gateway ARN arn:aws:ec2:us-east-1:307619016552:natgateway/nat-13980b9f202e1ac0a	Connectivity type Public	Created Monday, March 9, 2026 at 10:38:18 G MT+5:30	Deleted -
VPC vpc-0cf6c70159c65ea7f StartUp-VPC	Method of EIP allocation Automatic		

Step 4: Created the Route table for public path and as well as private path and added the root route path and associate the subnets

The screenshot displays the AWS Management Console interface for the 'Route tables' section. The left-hand navigation pane shows the 'VPC dashboard' with a sidebar for 'Virtual private cloud' (containing 'Your VPCs', 'Subnets', 'Route tables', 'Internet gateways', 'Egress-only internet gateways', 'Carrier gateways', 'DHCP option sets', 'Elastic IPs', 'Managed prefix lists', 'NAT gateways', 'Peering connections', and 'Route servers') and 'Security' (containing 'Network ACLs' and 'Security groups'). The main content area is titled 'Route tables (1/11) Info' and includes a search bar and a table of route tables. The table has columns for Name, Route table ID, Explicit subnet associations, Edge associations, Main, and VPC. The 'StartUp-Private-RouteTable' is selected, showing its details in the 'Routes' tab. The routes table shows two entries: a default route (0.0.0.0/0) pointing to a NAT gateway (nat-13980b9f202e1ac0a) and a local route (10.0.0.0/16). The 'Subnet associations' tab shows the route table is associated with the 'rtb-0d4f0bae70a1a7b57' subnet.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0ff830c13f9c73afe	-	-	Yes	vpc-0cf6c70159c65ea7f Star
Work Public Route Table	rtb-03c5b8ed8c32e716a	subnet-060edaf07ff8520...	-	No	vpc-0245f8065100bccbc Wo
-	rtb-0fd30cce7aba7404b	-	-	Yes	vpc-0b53dba63d46e9fc4
StartUp-Private-RouteTable	rtb-0d4f0bae70a1a7b57	subnet-002e3f6188c269f...	-	No	vpc-0cf6c70159c65ea7f Star
my-public-route-table-01	rtb-0accd5e512ea3e28f	subnet-084a4d9cff31287...	-	No	vpc-0469678595a1c1cbe my

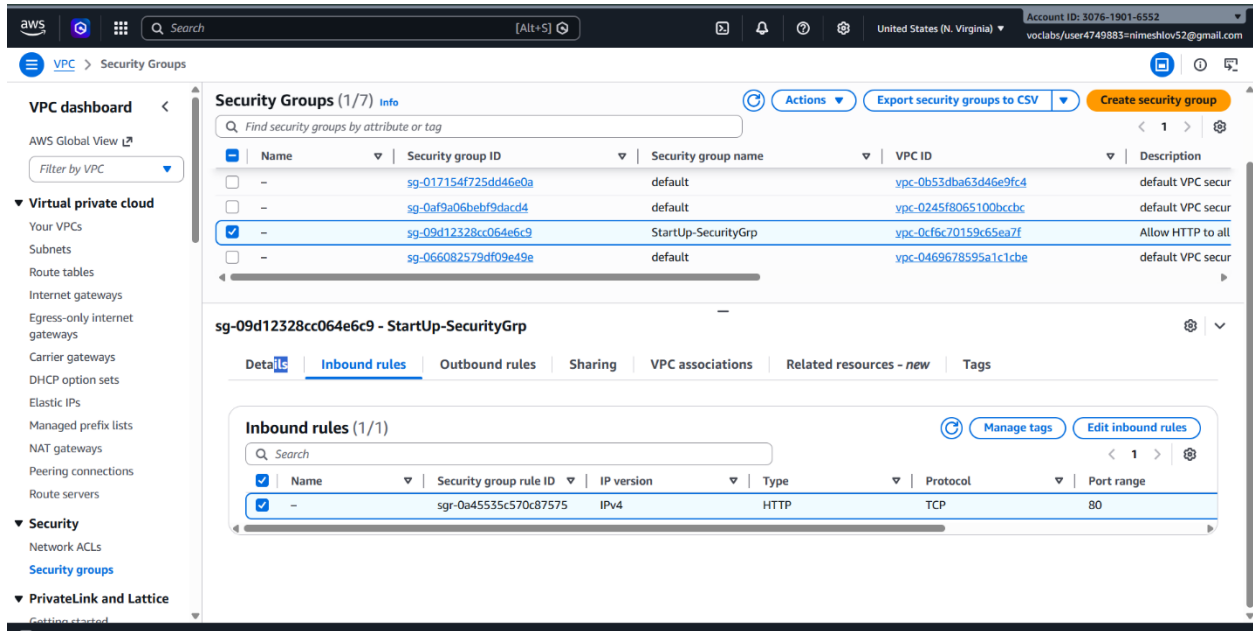
Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-13980b9f202e1ac0a	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

The screenshot displays the AWS Management Console interface for the 'Route tables' section, showing the configuration of a public route table. The left-hand navigation pane is identical to the previous screenshot. The main content area is titled 'Route tables (1/11) Info' and shows a table of route tables. The 'StartUp-Public-RouteTable' is selected, showing its details in the 'Routes' tab. The routes table shows two entries: a default route (0.0.0.0/0) pointing to an Internet Gateway (igw-09757064680bd0c44) and a local route (10.0.0.0/16). The 'Subnet associations' tab shows the route table is associated with the 'rtb-015c52ab93a1b8ad6' subnet.

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
-	rtb-0a20e05426306aa7f	-	nat-1207c8908050...	No	vpc-0469678595a1c1cbe my
my-private-route-table	rtb-084793ce1dfbca47a	-	-	No	vpc-0469678595a1c1cbe my
StartUp-Public-RouteTable	rtb-015c52ab93a1b8ad6	subnet-0ed5b3a3f18768...	-	No	vpc-0cf6c70159c65ea7f Star
-	rtb-0ff830c13f9c73afe	-	-	Yes	vpc-0cf6c70159c65ea7f Star
Work Public Route Table	rtb-03c5b8ed8c32e716a	subnet-060edaf07ff8520...	-	No	vpc-0245f8065100bccbc Wo

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-09757064680bd0c44	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Step 5: Created the security Group

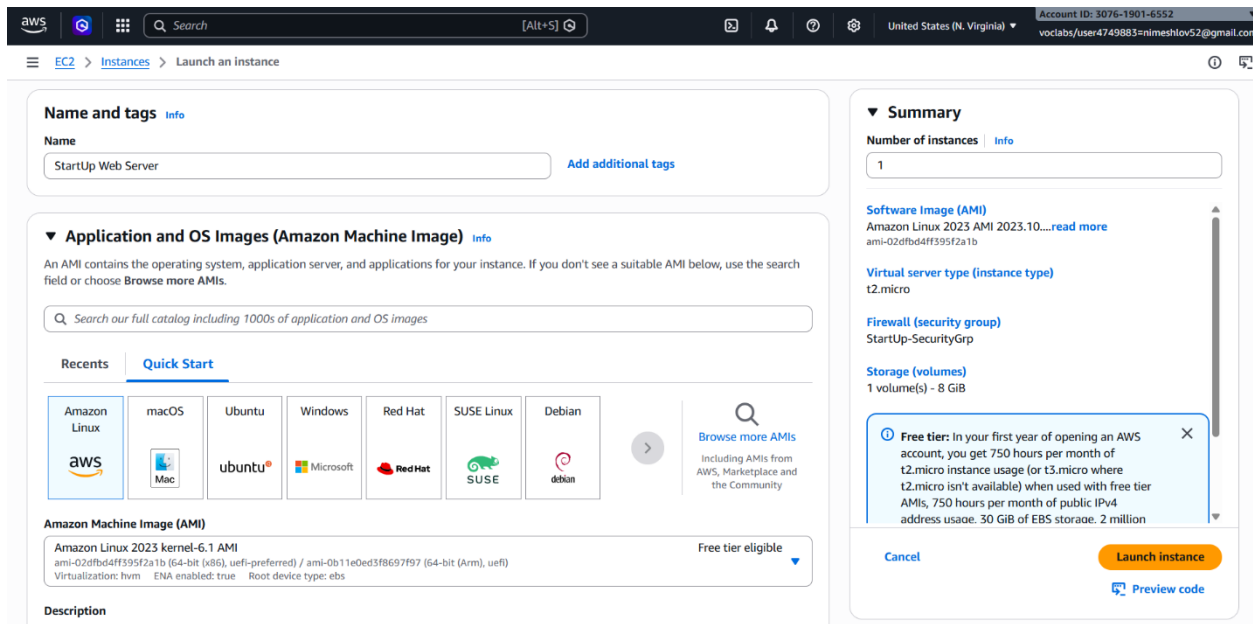


The screenshot shows the AWS Management Console interface for the 'Security Groups' page. The left sidebar contains navigation links for VPC, Virtual private cloud, and Security. The main content area displays a list of security groups. The 'StartUp-SecurityGrp' is selected, and its details are shown, including inbound rules. The inbound rules table shows a single rule for HTTP traffic on port 80.

Name	Security group ID	Security group name	VPC ID	Description
-	sg-017154f725dd46e0a	default	vpc-0b53dba63d4d6e9fc4	default VPC secur
-	sg-0af9a06bebf9dadc4	default	vpc-0245f8065100bcabc	default VPC secur
☒	sg-09d12328cc064e6c9	StartUp-SecurityGrp	vpc-0cf6c70159c65ea7f	Allow HTTP to all
-	sg-066082579df09e49e	default	vpc-0469678595a1c1cbe	default VPC secur

Name	Security group rule ID	IP version	Type	Protocol	Port range
☒	sg-0a45535c570c87575	IPv4	HTTP	TCP	80

Step 6: Created the Instance and configured with custom network configuration



The screenshot shows the AWS Management Console interface for the 'Launch an instance' page. The 'Name and tags' section shows the instance name 'StartUp Web Server'. The 'Application and OS Images (Amazon Machine Image)' section shows the 'Amazon Linux 2023 kernel-6.1 AMI' selected. The 'Summary' section shows the instance configuration, including the software image, virtual server type, firewall, and storage.

Name and tags

Name: StartUp Web Server

Application and OS Images (Amazon Machine Image)

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Recents: Quick Start

Amazon Linux, macOS, Ubuntu, Windows, Red Hat, SUSE Linux, Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02dfbd4ff395f2a1b (64-bit (x86), uefi-preferred) / ami-0b11e0ed3f8697797 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Summary

Number of instances: 1

Software Image (AMI): Amazon Linux 2023 AMI 2023.10...read more
ami-02dfbd4ff395f2a1b

Virtual server type (instance type): t2.micro

Firewall (security group): StartUp-SecurityGrp

Storage (volumes): 1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage. 30 GiB of EBS storage. 2 million

Launch instance

Account ID: 3076-1901-6552
voclabs/user4749883=nimeshlov52@gmail.com

United States (N. Virginia)

Search

[Alt+S]

EC2 > Instances > Launch an instance

▼ Network settings Info

VPC - required Info

vpc-0cf6c70159c65ea7f (StartUp-VPC)
10.0.0.0/16

Subnet Info

subnet-002e3f6188c269f01
VPC: vpc-0cf6c70159c65ea7f Owner: 307619016552 Availability Zone: us-east-1b (use 1-az1)
Zone type: Availability Zone IP addresses available: 251 CIDR: 10.0.2.0/24

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups Info

Select security groups

StartUp-SecurityGrp sg-09d12328cc064e6c9 X
VPC: vpc-0cf6c70159c65ea7f

Security groups that you add or remove here will be added to or removed from all your network interfaces.

► Advanced network configuration

▼ Summary

Number of instances Info

1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...read more
ami-02dfbd4ff395f2a1b

Virtual server type (instance type)

t2.micro

Firewall (security group)

StartUp-SecurityGrp

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage. 30 GiB of EBS storage. 2 million

Cancel Launch instance Preview code

Account ID: 3076-1901-6552
voclabs/user4749883=nimeshlov52@gmail.com

United States (N. Virginia)

Search

[Alt+S]

EC2 > Instances

EC2

Dashboard
AWS Global View
Events

▼ Instances

Instances
Instance Types
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Reserved Instances
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Capacity Reservations
Capacity Manager New

▼ Images

AMIs
AMI Catalog

▼ Elastic Block Store

Volumes
Snapshots
Lifecycle Manager

Instances (1/3) Info

Find Instance by attribute or tag (case-sensitive)

All states

Connect Instance state Actions Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	Bastion Host	i-0cbdbcc85e473b412	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	ec2-54-242
☒	StartUp Web S...	i-0fd60b5c5016a248a	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	ec2-44-222
☐	My Web Server	i-01d276609bf5f9761	Stopped	t3.micro	-	View alarms +	us-east-1a	-

i-0fd60b5c5016a248a (StartUp Web Server)

Details Status and alarms Monitoring Security Networking Storage Tags

▼ Instance summary Info

Instance ID

i-0fd60b5c5016a248a

IPv6 address

-

Public IPv4 address

44.222.157.243 | open address

Instance state

Running

Private IPv4 addresses

10.0.2.204

Public DNS

ec2-44-222-157-243.compute-1.amazonaws.com | open address